
AI-102 Natural Language Processing (NLP)

Dr. Jimmy Ardiansyah



EMERGING &
COLLABORATIVE STUDIES



A blue-tinted background image of a statue of Prometheus, a figure from Greek mythology, holding a torch. The statue is positioned in the center-right of the frame. In the top right corner, there are several parallel orange diagonal lines. A thin orange vertical line is on the left side of the slide, and a thin orange horizontal line is at the bottom.

Beyond Natural Language Processing

Responsible AI

- **Fairness:** avoid systematic harm to any group, check and reduce bias
- **Transparency:** explain what the system does, when AI is used, and its limits
- **Accountability:** humans remain responsible for AI decisions and outcomes
- **Privacy:** collect minimal data, protect it, and respect user consent
- **Safety and robustness:** design for failures, adversarial inputs, and misuse
- **Human centered:** AI should support human judgment, not quietly replace it



MLOps and LLMOps

MLOps is the practice of applying DevOps principles to machine learning models to streamline their development, deployment, and monitoring.

LLMOps is a specialized subset of MLOps specifically for large language models, addressing their unique requirements like huge size, complex training, and different monitoring needs.

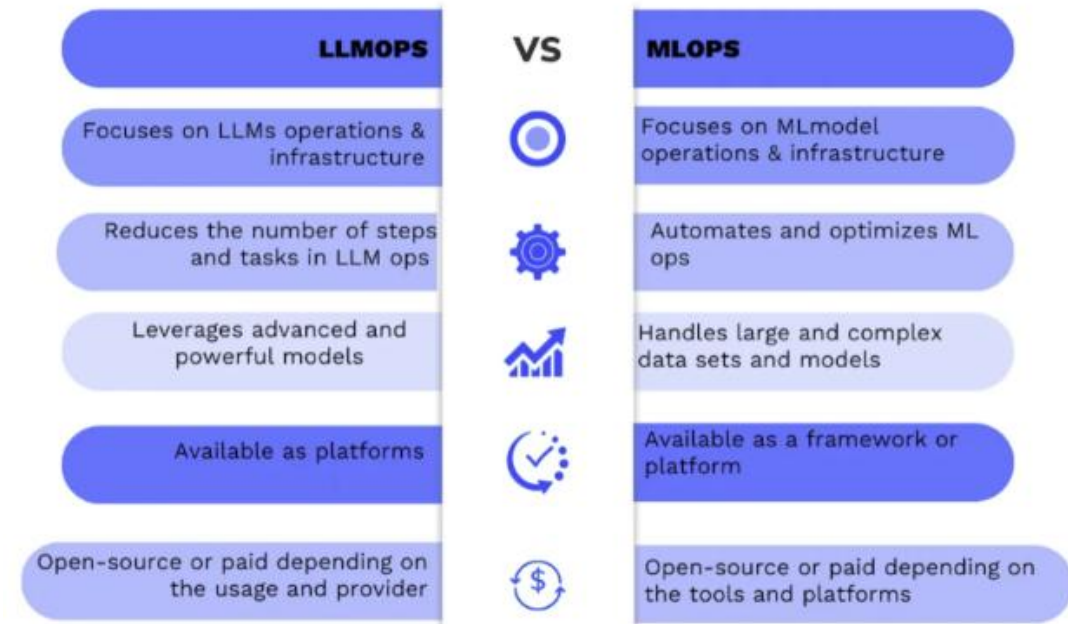
Both aim to make AI systems reliable, but MLOps focuses on general predictive models, while LLMOps tackles generative models and their specific challenges such as hallucination detection and prompt injection security

MLOps (Machine Learning Operations) key activities:

- Experiment tracking
- Feature engineering
- Model training and evaluation
- Performance monitoring for data drift
- Deployment through pipelines and containerization

LLMOps (Large Language Model Operations) key activities:

- Fine-tuning and prompt engineering
- Human feedback
- Specialized monitoring
- External model integration

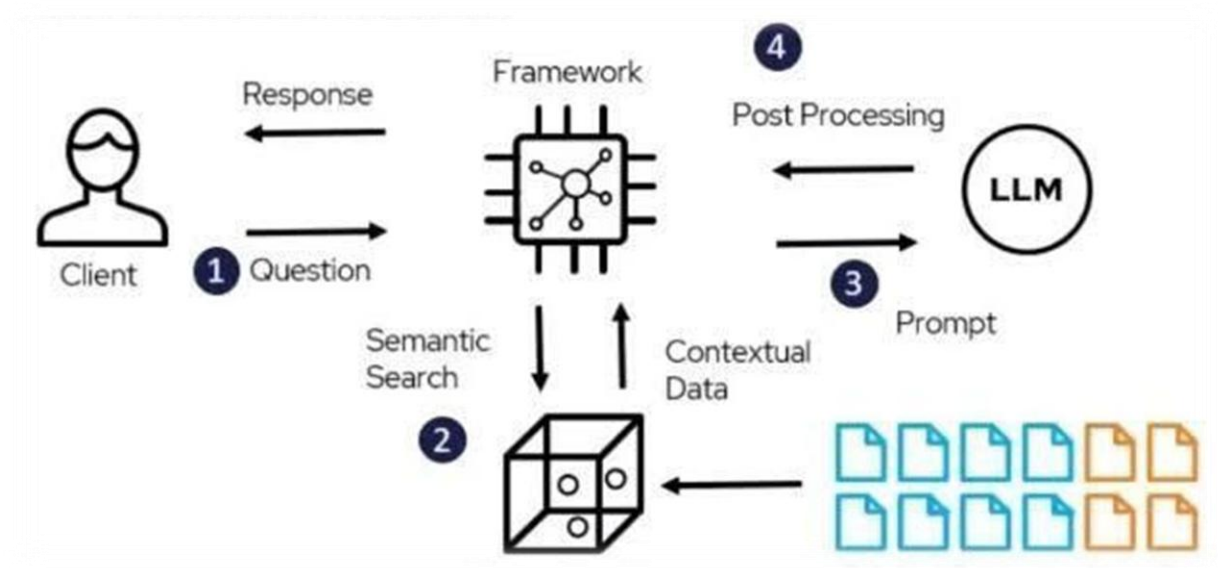


Retrieval Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is an artificial intelligence (AI) framework that enhances the accuracy and reliability of large language models (LLMs) by connecting them to external, up-to-date knowledge sources before generating a response. This approach allows the AI to use specific, verified information rather than relying solely on its static, pre-trained data

RAG operates in two primary phases: retrieval and generation.

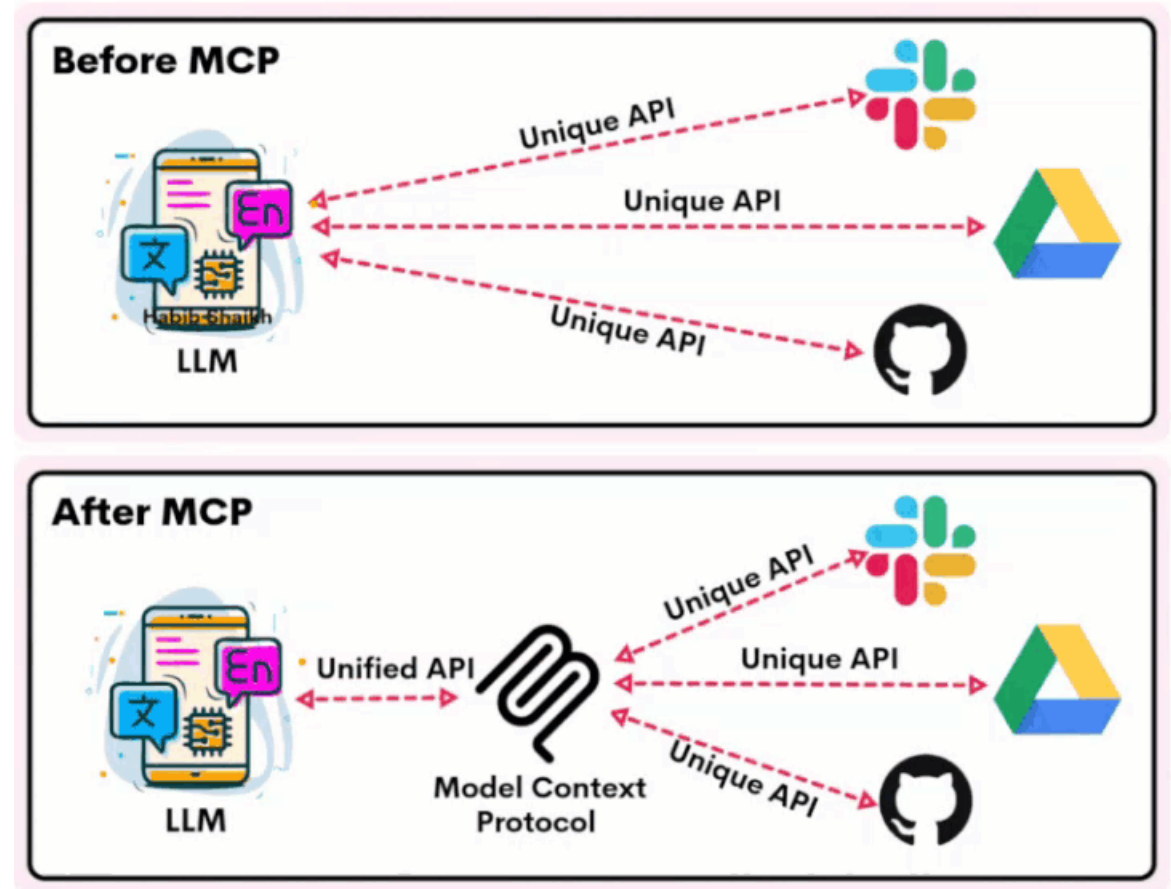
- Ingestion & Storage: Collect external data (docs, DBs, web pages), split into chunks, turn into embeddings, and store in a vector database.
- Retrieval: When the user asks a question, run a semantic search to pull the most relevant chunks.
- Augmentation: Combine the user's question with the retrieved chunks to form an augmented prompt.
- Generation: Send this augmented prompt to the LLM, which uses the added context plus its training to produce an accurate, context-aware answer.



Model Context Protocol (MCP)

The Model Context Protocol (MCP) is an open-source, client-server communication standard that allows LLMs to connect with and use external data sources and tools in a standardized way

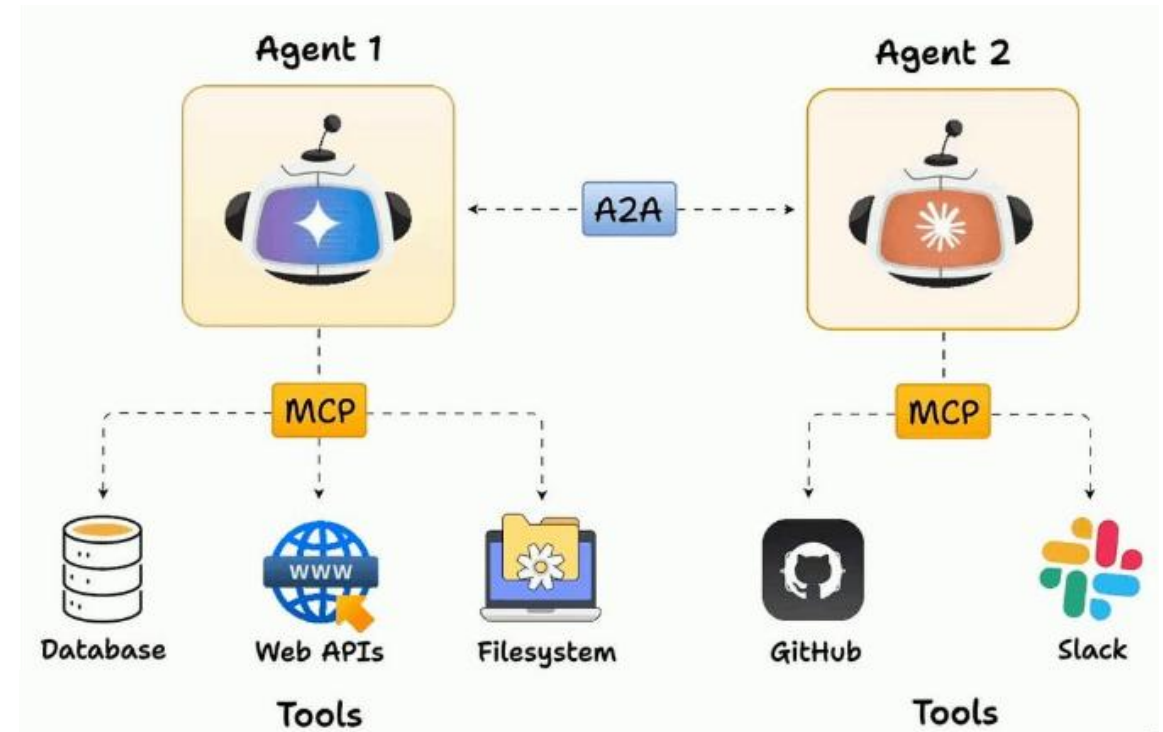
- **Enhances AI capabilities:** Allows AI agents to access real-time data and perform actions. **Reduces "hallucinations":** Provides AI models with access to reliable, verified external data, reducing the likelihood of them generating plausible-sounding but incorrect information.
- **Simplifies development:** Replaces the need for custom, point-to-point integrations for every AI model and external system with a single, reusable protocol.
- **Improves security:** Creates a structured, controlled interface for AI access, defining exactly what an agent can and cannot do, thus preventing data spillage and unauthorized actions.



Agent to Agent Protocol (A2A)

An **Agent-to-Agent (A2A) protocol** is an open communication standard that enables different artificial intelligence (AI) agents, regardless of their underlying frameworks or vendors, to discover each other's capabilities, exchange information, delegate tasks, and collaborate to achieve complex goals. It provides a universal language for agent interoperability, moving beyond isolated agent systems to an interconnected ecosystem

Model Context Protocol (MCP): While A2A focuses on agent-to-agent collaboration, MCP standardizes how a single agent connects to its tools, data sources, and external APIs. An agent might use MCP to check an inventory database and then use A2A to communicate with a supplier agent to place an order





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